PFUMA

Digitizing Zimbabwe's Livestock Trade — a Five-Role Platform for Disease Control, Anti-Theft Verification & Market Access

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Introduction

Livestock trade in Zimbabwe runs on paper, word-of-mouth, and informal trust. January Disease (Theileriosis) alone has killed over 500,000 cattle since 2016. Stock theft is organized enough that a single recent police crackdown produced over 3,400 arrests. Communal farmers — the majority — are routinely excluded from formal credit and verified trade because signup and paperwork assume freehold title they don't hold.

Objective

Build and deploy a role-aware digital platform — Farmer, Veterinarian, Supplier, Retailer, Police — that operationalizes Zimbabwe's actual legal framework for livestock trade (Stock Theft Prevention Act, Animal Health Act, Veterinary Surgeons Act, Consumer Protection Act) as working software, not an afterthought bolted onto a generic farm app.

Methodology

A React (Vite) web app and Expo mobile app share a Flask/MySQL backend with JWT authentication, bcrypt password hashing, and per-owner data scoping. An AI compliance assistant (Jinda) answers legal/health questions from researched Zimbabwean law and refuses to leak other users' data. ESP32/LoRa IoT hardware pairs collars and base stations to a farmer's account for herd health and anti-theft monitoring. The system architecture is shown at right — note the Police Clearance Layer.

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stakeholder roles	backend API endpoints	Zimbabwean law/species docs researched

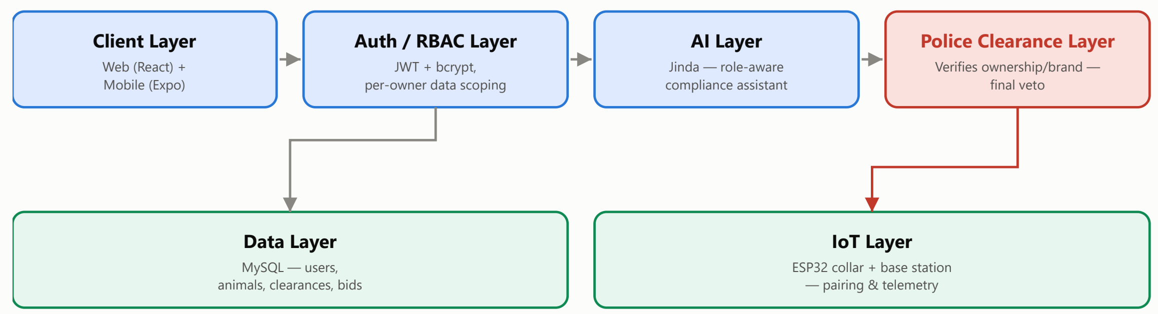
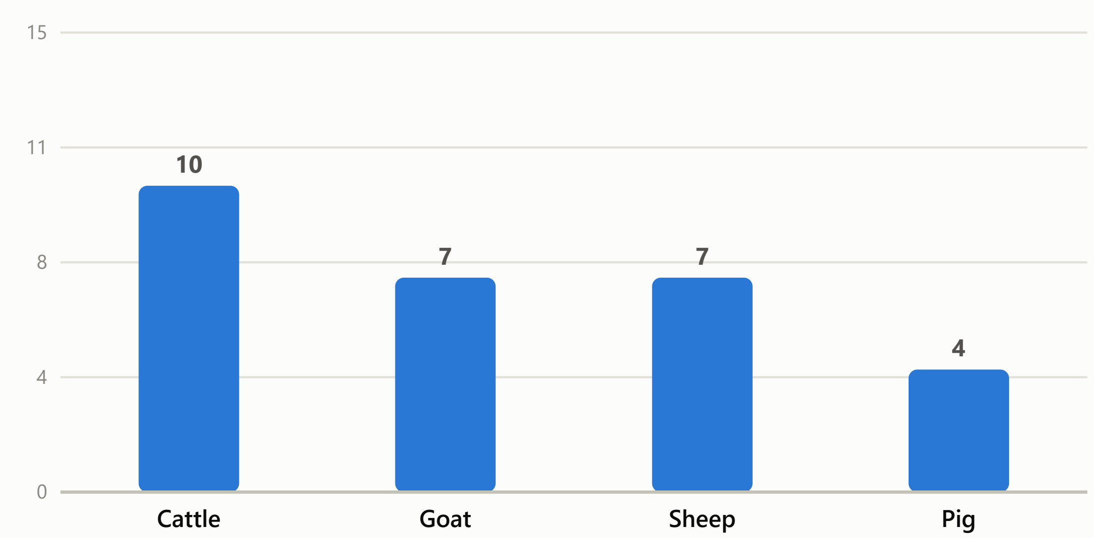


Figure 1. System architecture — the Police Clearance Layer always has final veto over every livestock listing, mirroring Zimbabwe's real ownership → vet → police clearance sequence.

Disease Coverage by Species

15 notifiable/high-impact diseases modeled in the diagnostics engine, spanning WOAH-notifiable and Zimbabwe-endemic conditions (FMD, January Disease, Anthrax, ASF, PPR, CCPP, and more).



Results

Every claim below was verified by exercising the running system (curl against the live Flask API, database round-trips), not asserted from the code alone. Rather than present the platform as uniformly "done," each capability is labelled honestly:

CAPABILITY	STATUS
Auth (bcrypt + JWT), per-role & per-owner data scoping	Real
Police signup review + sale-clearance workflow	Real
AI compliance assistant, role-aware data guard	Real
IoT device pairing + telemetry intake endpoint	Real
Live IoT sensor dashboard (continuous reading history)	Simulated

Analysis

The Police Clearance Layer is the structural core of the platform, not a UI badge: a livestock listing is created with status `pending_clearance` and is database-enforced invisible to buyers (`WHERE status='available'`) until an officer resolves it. This mirrors the same "software assists, but a hard rules layer has final veto" pattern used in Team GRIWD's MASAISAI submission to the POTRAZ AI4I challenge — disease/AI signals inform the decision, but only a verified authority can grant access.

Conclusion

A working, tested prototype demonstrates that a legally-grounded, role-aware livestock platform is achievable today — not a future roadmap item — giving Zimbabwe's Agricultural Show a concrete technical case for formal, law-compliant digital trade infrastructure that protects farmers, deters stock theft, and supports national disease surveillance.

Key Sources & Acknowledgements

Stock Theft Prevention Act [Chapter 9:18], Zimbabwe

Animal Health Act [Chapter 19:01] & Animal Health (General) Regulations, 1994

Veterinary Surgeons Act [Chapter 27:15] — Council of Veterinary Surgeons of Zimbabwe (CVSZ)

Consumer Protection Act [Chapter 14:44], Zimbabwe, 2019

World Organisation for Animal Health (WOAH) & FAO disease references

Zimbabwe Republic Police — Anti Stock Theft Unit statements

Full legal research & source citations: /compliance folder in repository

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